

6. Specialized Instrument and Devices

Objectives

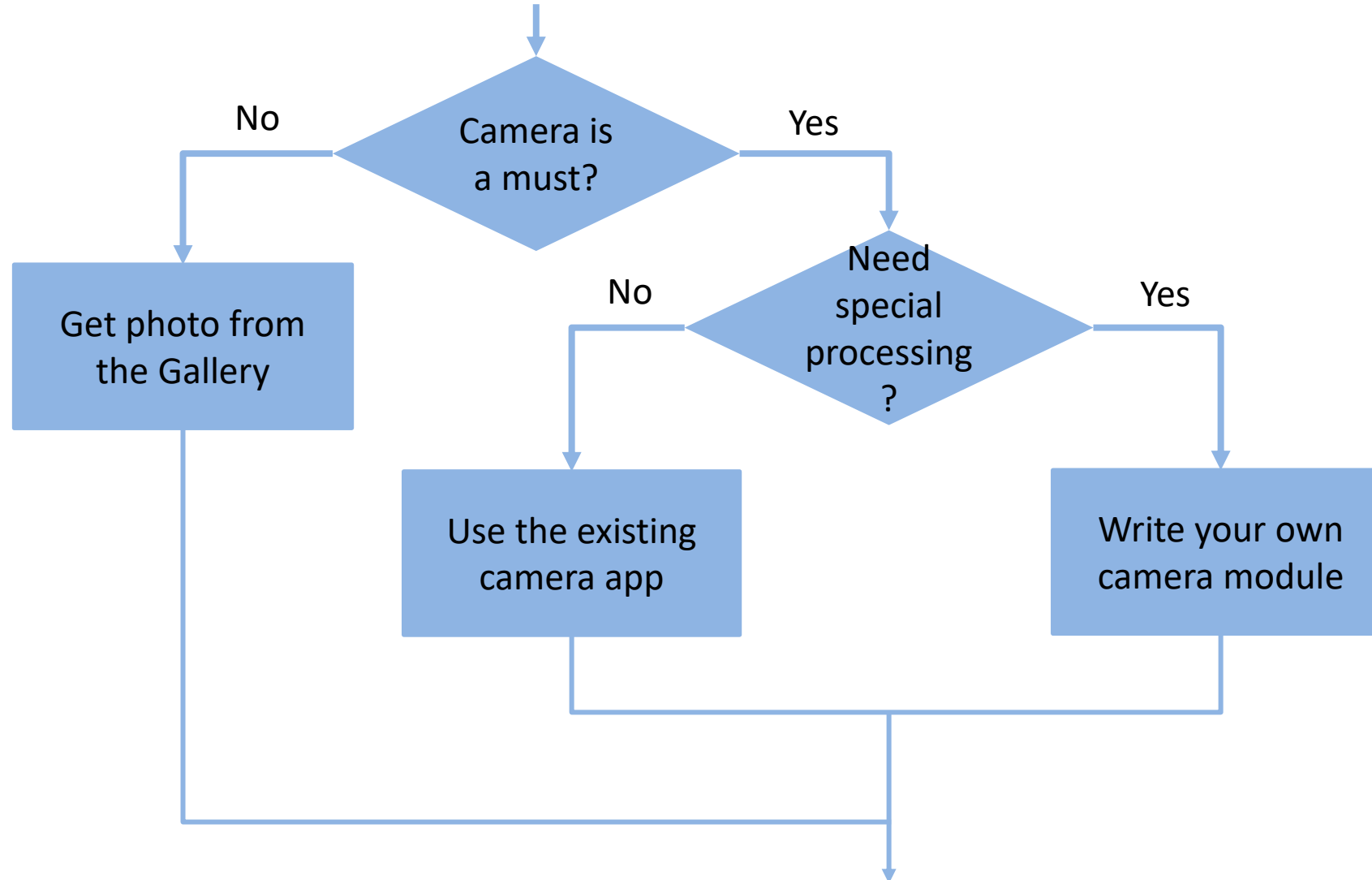
Explain basic requirements of a camera app

Identify and explain other hardware sensors

Develop app that could capture image and video

Implement app that can play audio and video files

Camera - Considerations



Methods

Get from the Gallery

```
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
```

```
implementation("io.coil-kt:coil-compose:2.4.0") // Or latest version
```

```
val context = LocalContext.current  
val selectedImageUri = remember { mutableStateOf<Uri?>(null) }
```

```
val launcher = rememberLauncherForActivityResult(  
    contract = ActivityResultContracts.GetContent()  
) { uri: Uri? ->  
    selectedImageUri.value = uri  
}
```

Methods - Get from the Gallery

```
Column(  
    modifier = Modifier  
        .fillMaxSize()  
        .padding(16.dp),  
    horizontalAlignment = Alignment.CenterHorizontally  
) {  
    Button(onClick = {  
        launcher.launch("image/*") // Opens gallery to select image  
    }) {  
        Text("Pick Image from Gallery")  
    }  
  
    Spacer(modifier = Modifier.height(20.dp))  
  
    selectedImageUri.value?.let { uri ->  
        Image(  
            painter = rememberAsyncImagePainter(uri),  
            contentDescription = null,  
            modifier = Modifier  
                .size(200.dp)  
                .clip(RoundedCornerShape(8.dp))  
        )  
    }  
}
```

Methods

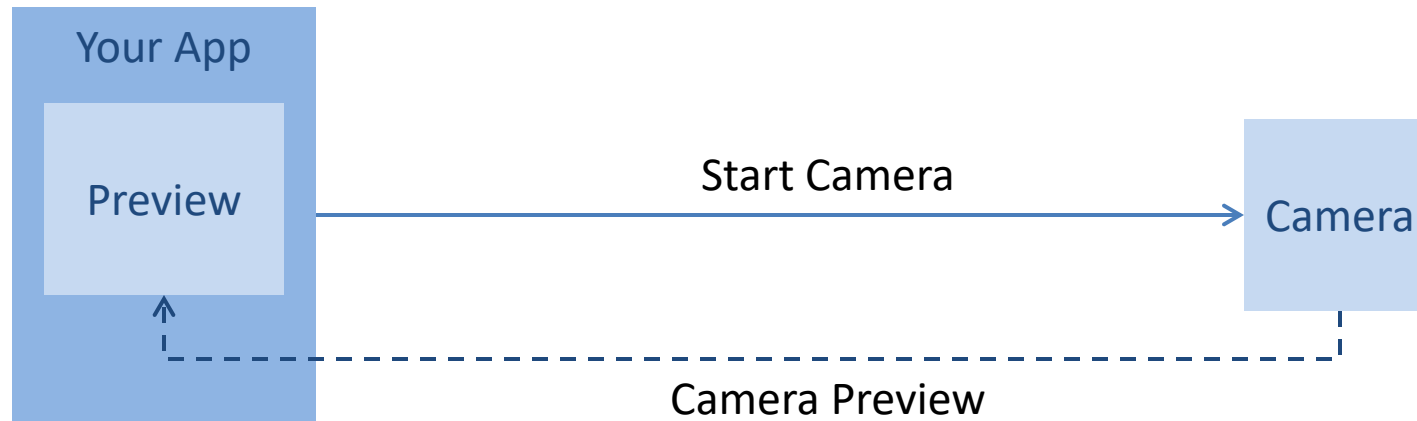
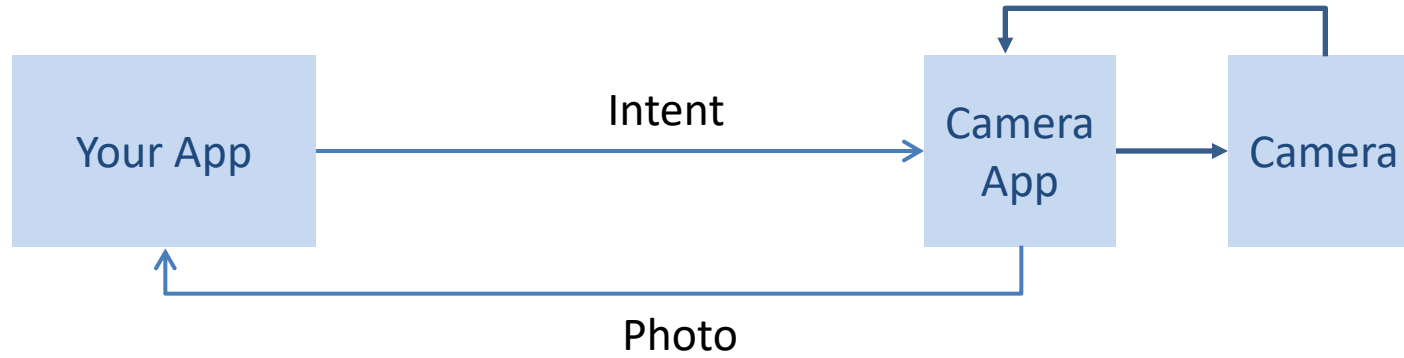
Use an existing camera app

- Simple implementation - using Intent Class
- Obtain a photo from the camera app
- No need permission

Build your own camera function

- Complex implementation - write your own code using CameraX or Camera2 APIs
- Create a camera preview
- Request Camera permission

The Basics



Manifest Declarations

Camera Features

- Prevents your app from being installed to devices that do not include a camera

```
<uses-feature android:name="android.hardware.camera" />
```

Manifest Declarations

3. Storage Permission

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"  
    android:maxSdkVersion = "18" />
```

4. Audio Recording Permission

```
<uses-permission android:name="android.permission.RECORD_AUDIO" />
```

5. Location Permission

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
```

Using Camera Intent

A quick way to enable taking pictures or videos in your app with very minimum coding efforts

Process:

1. Create an Intent using `MediaStore.ACTION_IMAGE_CAPTURE` or `MediaStore.ACTION_VIDEO_CAPTURE`
2. Execute the Intent using `startActivity()`
3. Set up an `onActivityResult()` to receive Intent result

Take a photo

```
val context = LocalContext.current
val imageUri = remember { mutableStateOf<Uri?>(null) }

// Create temp file for storing the captured image
val photoFile = remember {
    File.createTempFile("photo_", ".jpg", context.cacheDir).apply {
        createNewFile()
        deleteOnExit()
    }
}

val photoUri = FileProvider.getUriForFile(
    context,
    "${context.packageName}.fileprovider",
    photoFile
)

val cameraLauncher = rememberLauncherForActivityResult(
    contract = ActivityResultContracts.TakePicture()
) { success ->
    if (success) {
        imageUri.value = photoUri
    }
}
```

What can your phone camera do?

Claim vehicle insurance

Cheque deposit

Search

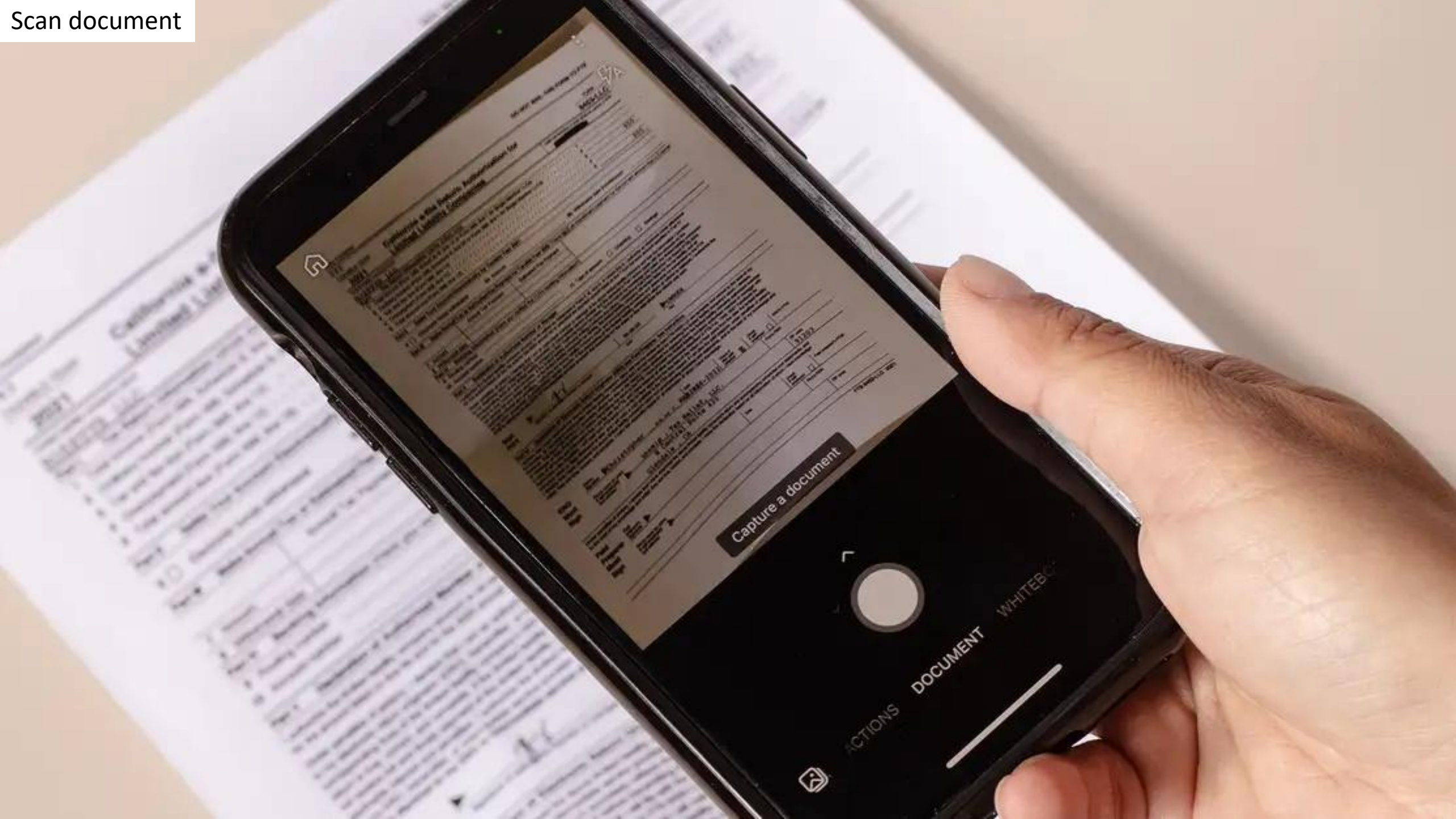
மொழிபெயர்

Eye test



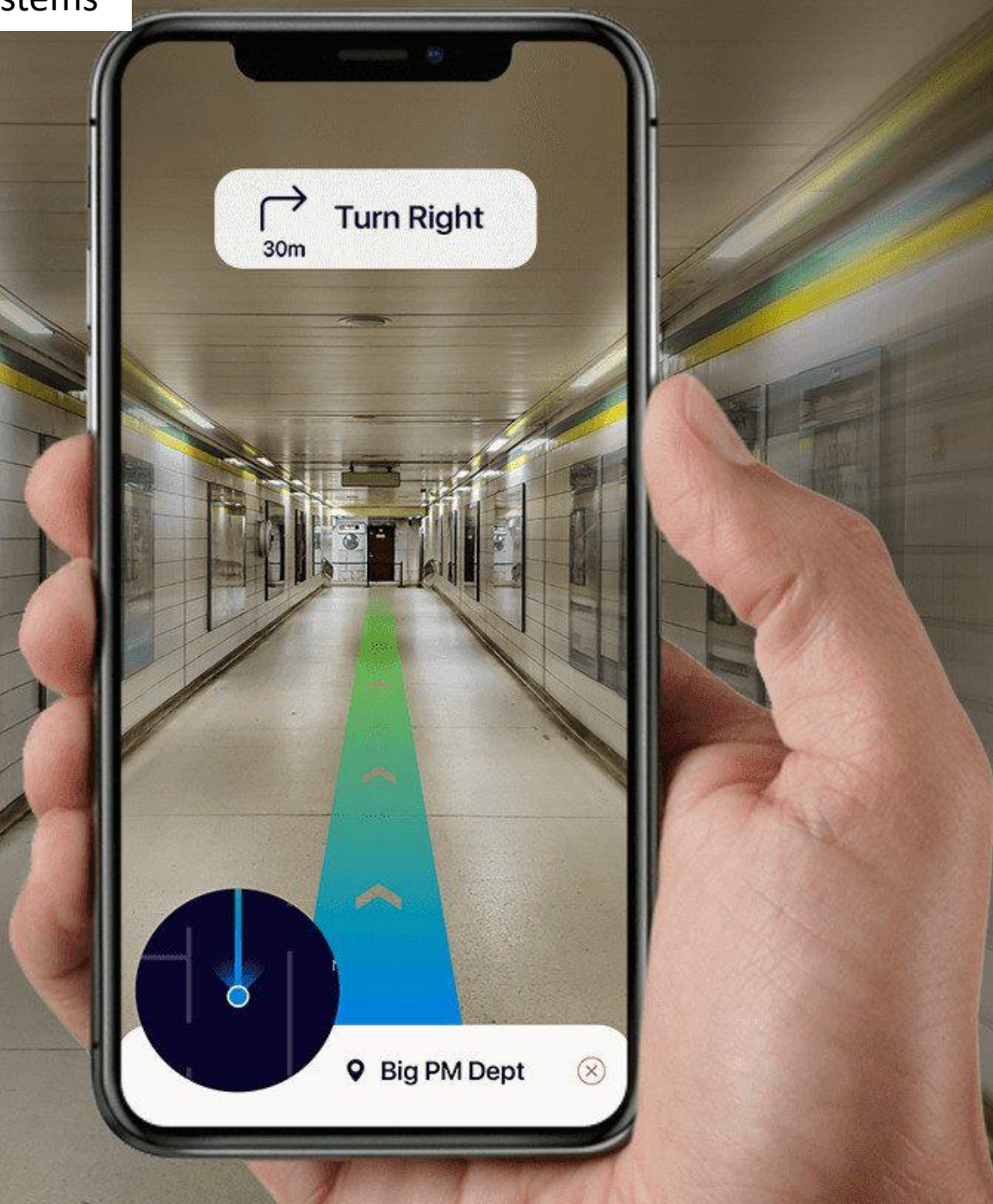
<https://developers.google.com/ml-kit/vision/text-recognition>

Scan document



Recognize object and text





Decode a scaled image

App may running out of memory after displaying too many images

Images should be scaled to match the size of the destination view

Decode a scaled image

```
fun LoadScaledImage(uri: Uri) {  
    val context = LocalContext.current  
    val bitmap by remember(uri) {  
        mutableStateOf(  
            decodeScaledBitmapFromUri(context, uri, reqWidth = 600, reqHeight = 600)  
        )  
    }  
  
    bitmap?.let {  
        Image(  
            bitmap = it.asImageBitmap(),  
            contentDescription = null,  
            modifier = Modifier  
                .size(300.dp)  
                .clip(RoundedCornerShape(8.dp))  
        )  
    }  
}
```

Barcode Scanning

Machine Learning (ML) barcode scanning API

On device, no need a network

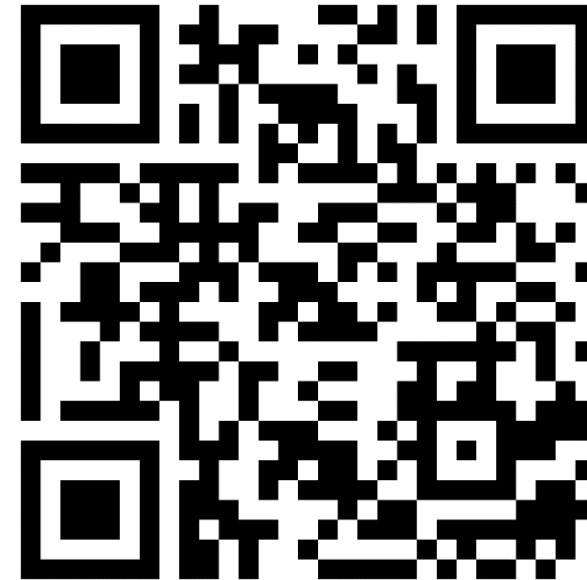
Support 2D format (QR Code)



<https://developers.google.com/ml-kit/vision/barcode-scanning>

6.1 Questions?

1. “You must obtain permission to enable the camera feature in your app.” Comment on this statement
2. How to prevent your app from being installed on devices that do not have a camera?
3. What is the quick way to enable taking pictures or videos in your app without a lot of extra code?



Audio Streams

Android maintains a separate audio stream for:

- playing music,
- alarms,
- notifications,
- the incoming call ringer,
- system sounds,
- in-call volume, and
- etc

Audio Stream

Most of these streams are restricted to system events

We focus on playing audio using the `STREAM_MUSIC` stream

Use the `STREAM_MUSIC` stream for: background music or sound effects

Audio Playing

Classes to play sound and video in the Android framework:

- MediaPlayer: plays sound and video.
- AudioManager: manages audio sources and audio output on a device.

MediaPlayer

Plays local and external (streaming) files

Supports any media codec that is provided by the Android platform and those that are device-specific

Recommendation: use core media formats

MediaPlayer

- Permission declaration

- If you are using MediaPlayer to stream network-based content

```
<uses-permission android:name="android.permission.INTERNET" />
```

- If your player application needs to keep the screen from dimming or the processor from sleeping

```
<uses-permission android:name="android.permission.WAKE_LOCK" />
```

Core Media Format

Audio: .3gp .mp3 .mp4 .mid .wav .ogg

Picture: .jpg .gif .png .bmp

Video: .3gp .mp4

Full list: <http://developer.android.com/guide/appendix/media-formats.html>

MediaPlayer

```
val context = LocalContext.current
var isPlaying by remember { mutableStateOf(false) }
val mediaPlayer = remember {
    MediaPlayer().apply {
        setDataSource(audioUrl)
        prepare() // Prepare synchronously (for local/fast streams)
    }
}

Button(onClick = {
    if (mediaPlayer.isPlaying) {
        mediaPlayer.pause()
        isPlaying = false
    } else {
        mediaPlayer.start()
        isPlaying = true
    }
}) {
    Text(if (isPlaying) "Pause" else "Play")
}
```

Releasing the MediaPlayer

A MediaPlayer can consume valuable system resources

Always call `release()` to make sure any system resources allocated to it are properly released

```
DisposableEffect(Unit) {  
    onDispose {  
        mediaPlayer.release() // Release when Composable leaves composition  
    }  
}
```

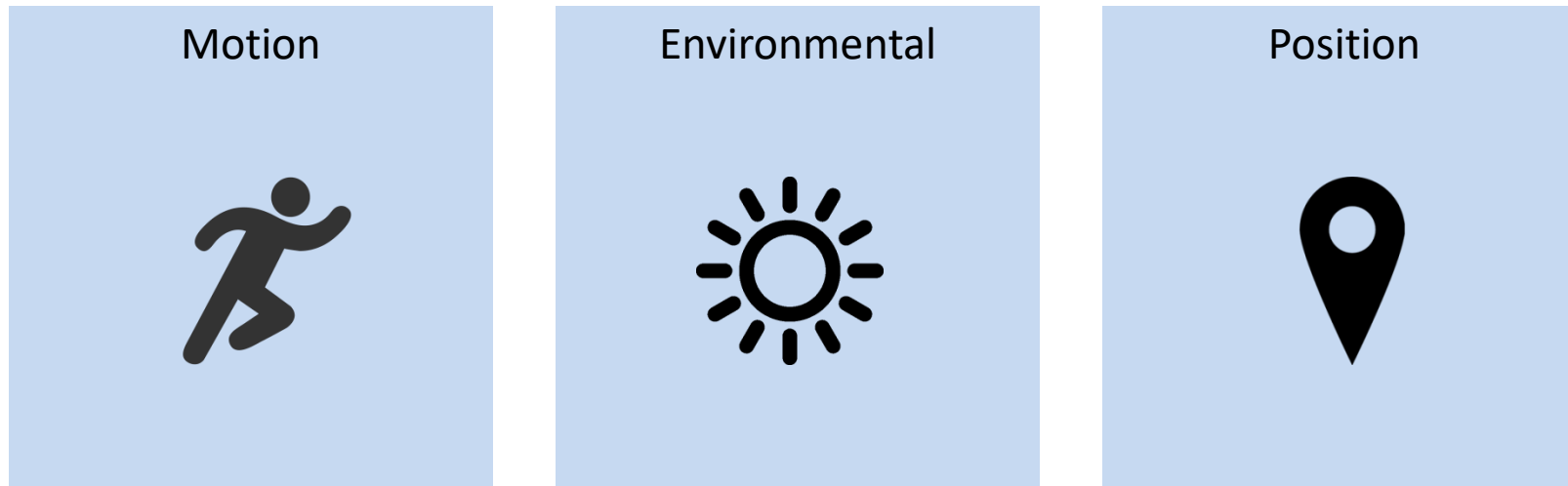
6.2 Questions

1. “You are free to use any media codec in your Android app”. Comment on this statement.
2. What is the purpose of the `android.permission.WAKE_LOCK` permission?
3. The ____ method is used to remove resources associated with a `MediaPlayer` object.



Sensors Overview

- Three categories





Motion sensors

Measure acceleration forces and rotational forces along three axes

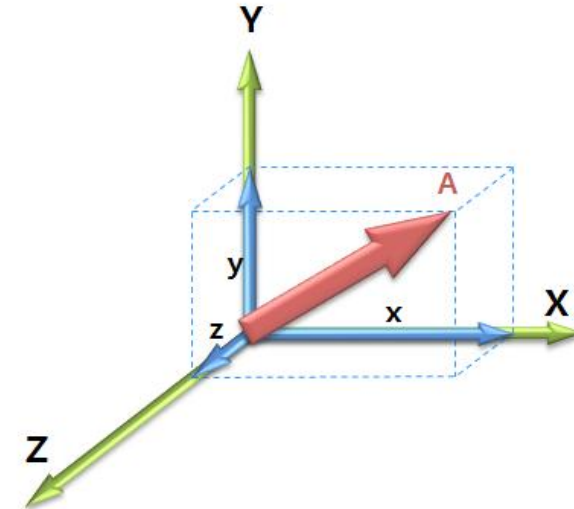
1. accelerometers
2. gravity sensors
3. gyroscopes
4. rotational vector sensors



Motion sensors

Accelerometer

- Provides data on the rate of change in **velocity**, excluding gravity
- It senses **axis orientation**

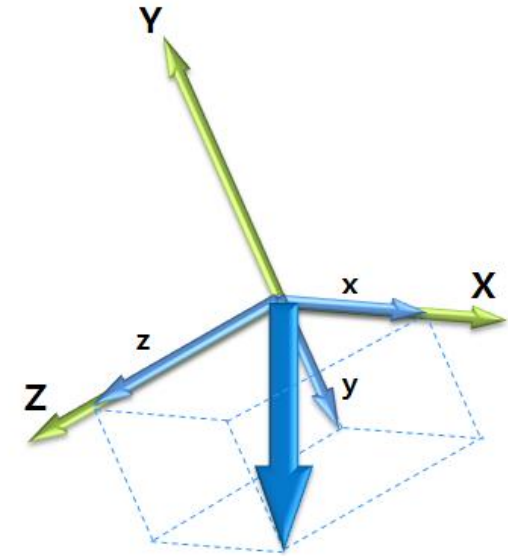
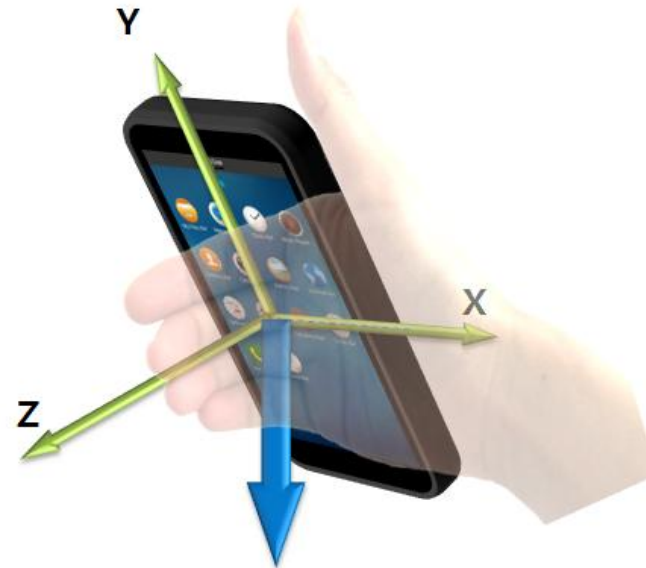




Motion sensors

Gravity sensor

- It measures the **direction** and intensity of **gravity**
- Checks the relative direction of a device within a space

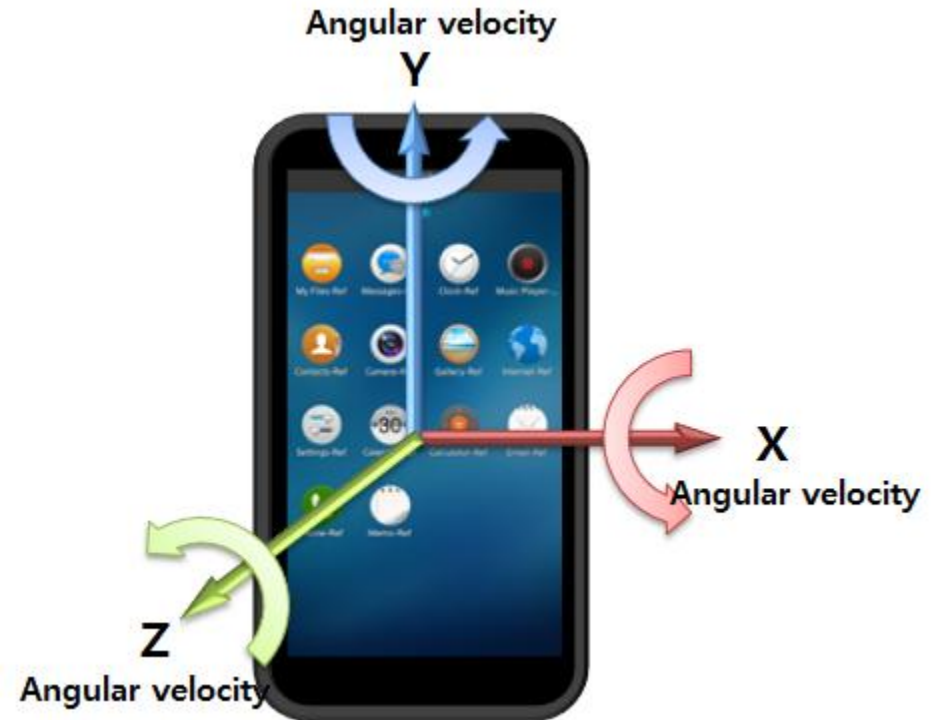




Motion sensors

Gyroscope

- It helps the **accelerometer** out with understanding which way your phone is **orientated**



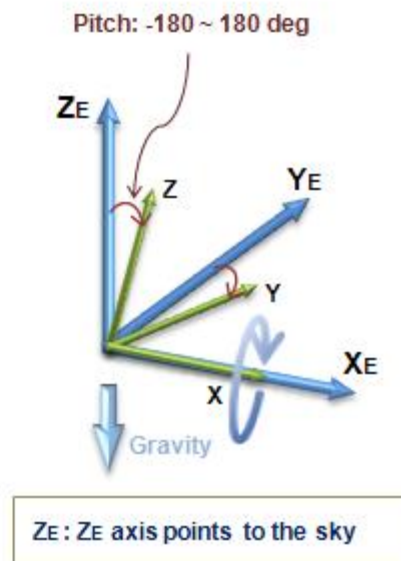
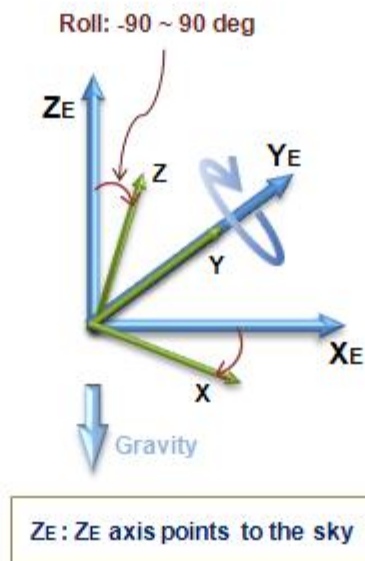
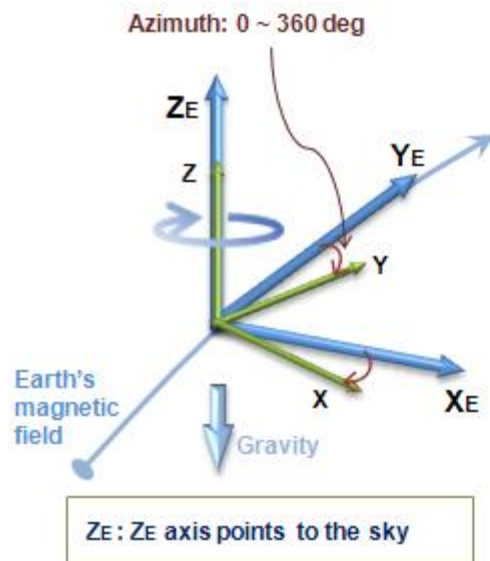
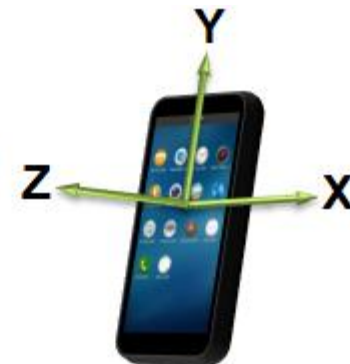


Motion sensors

Rotational vector sensors

- To monitor and measure **turning** movements

X, Y, Z: Frame fixed on the device
(Common for the frames below)





POS. **5/8**

DIST. **23%**

TOUCHDRIVE **ON**

KM/H **232**

0:25:24



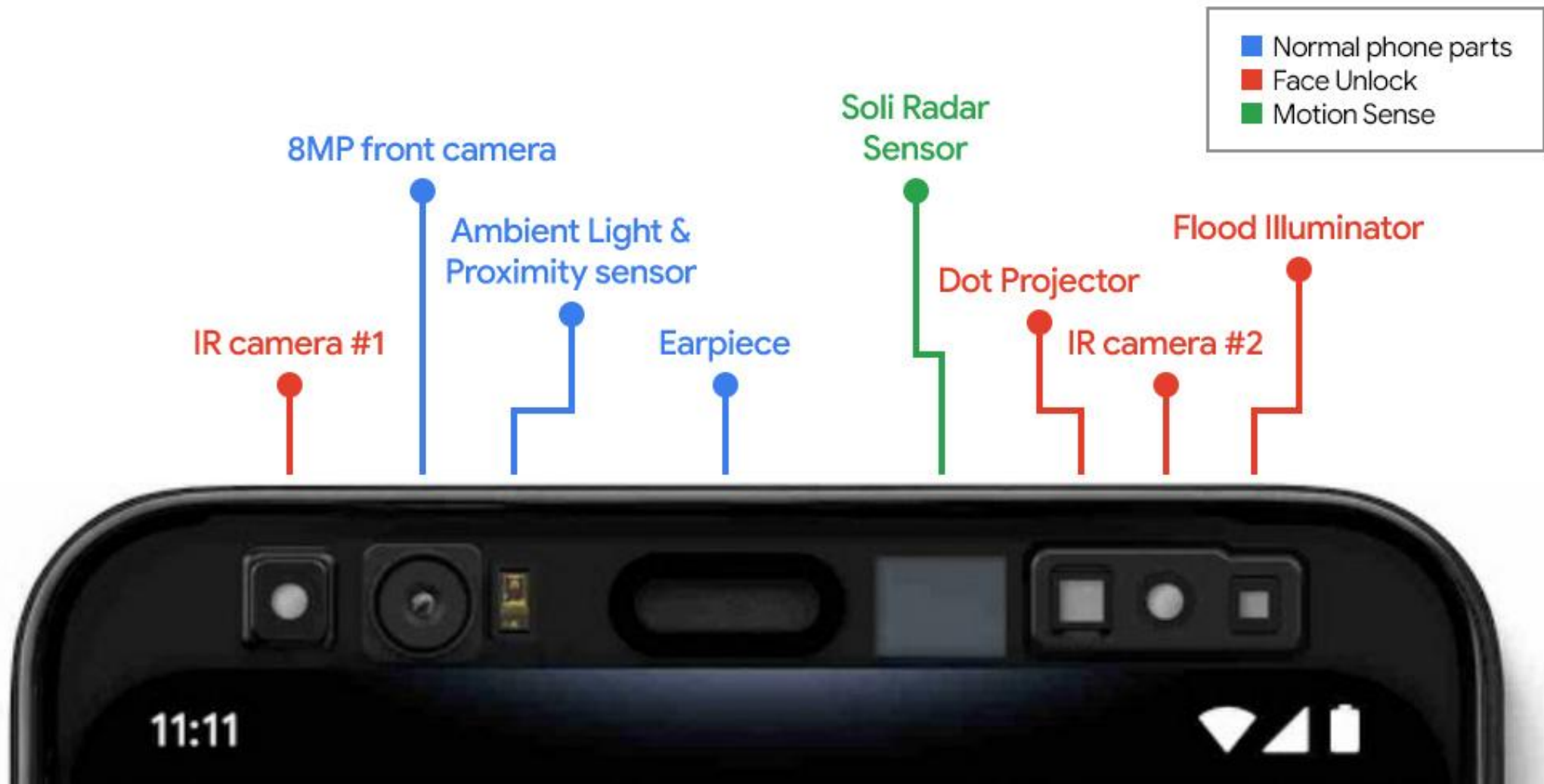




Environmental sensors

Measure various environmental parameters, such as ambient air temperature and pressure, illumination, and humidity **near an Android-powered device.**

1. barometers
2. photometers, and
3. thermometers



Ambient light sensor

Speaker

Proximity sensor

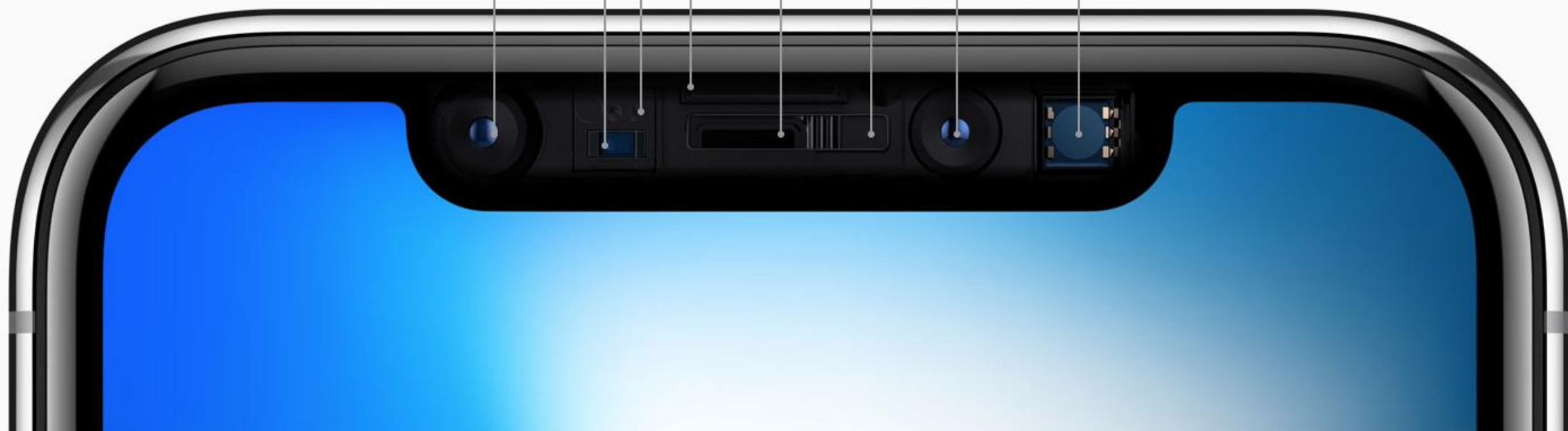
Microphone

Flood Illuminator

7MP camera

Infrared camera

Dot projector

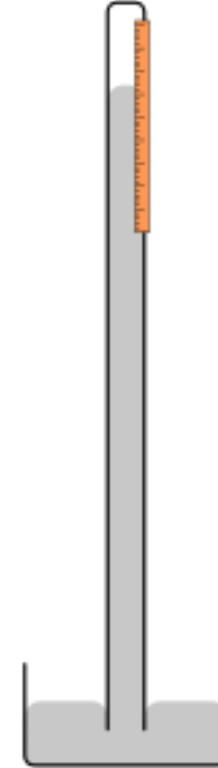




Environmental sensors

Barometer

- It measures **atmospheric pressure**
- It **assists the GPS** chip inside the device to get a faster lock by instantly delivering **altitude** data (vertical location).

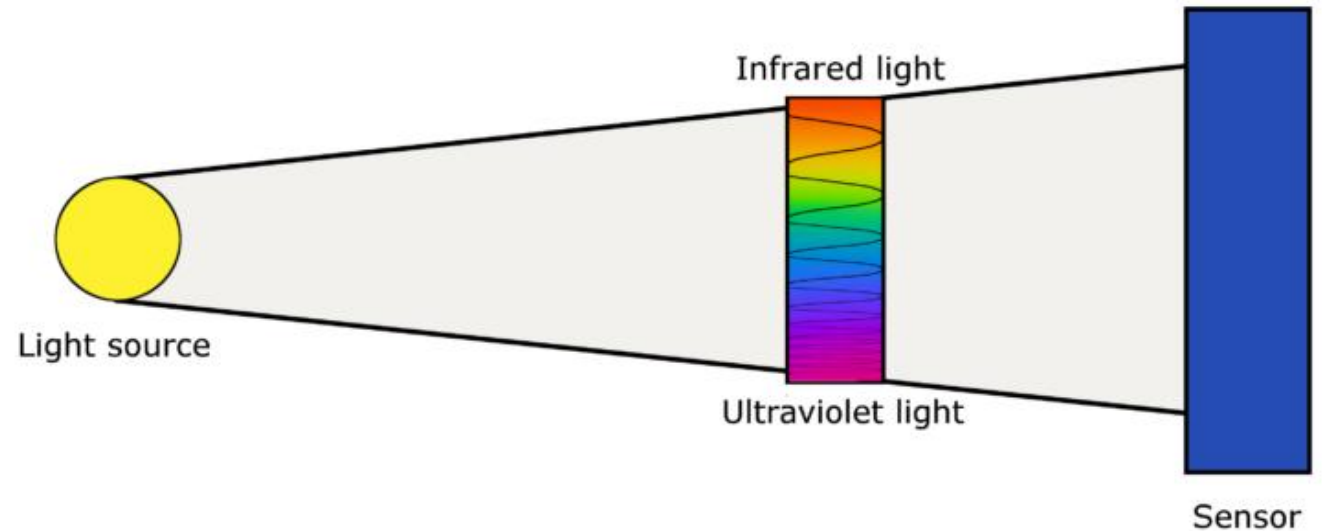




Environmental sensors

Photometer / Light sensor

- It senses the amount of **ambient light** present

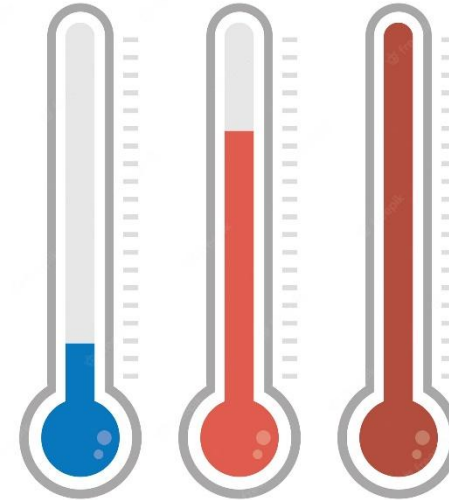




Environmental sensors

Thermometer

- It measures **temperature within a mobile device**
- To **avoid overheating** of the **internal components**





Position sensors

Measure the physical position of a device

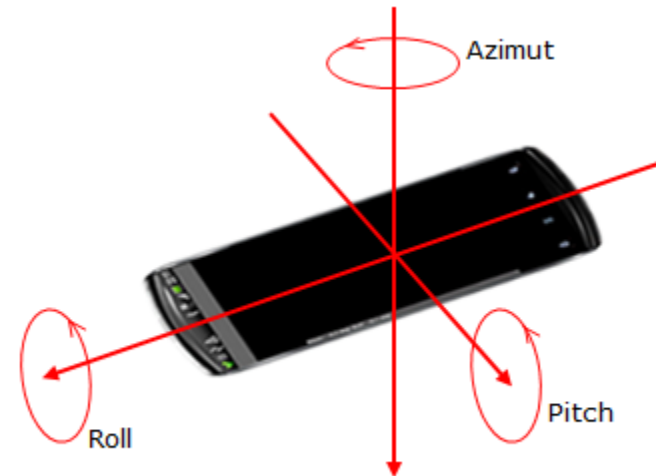
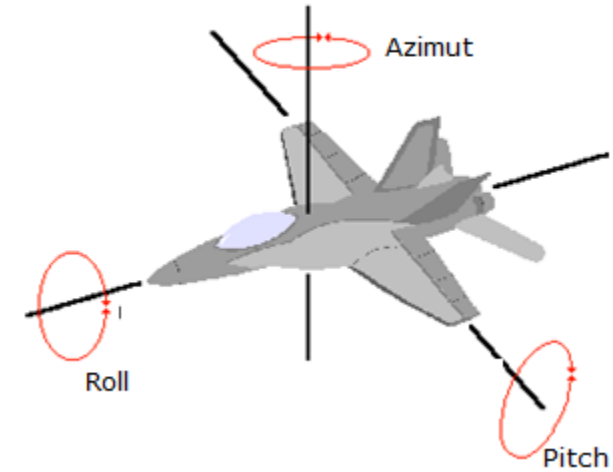
- orientation sensors
- magnetometers



Position sensors

Orientation sensors

- Measures the orientation of a device relative to an orthogonal coordinate frame





Position sensors

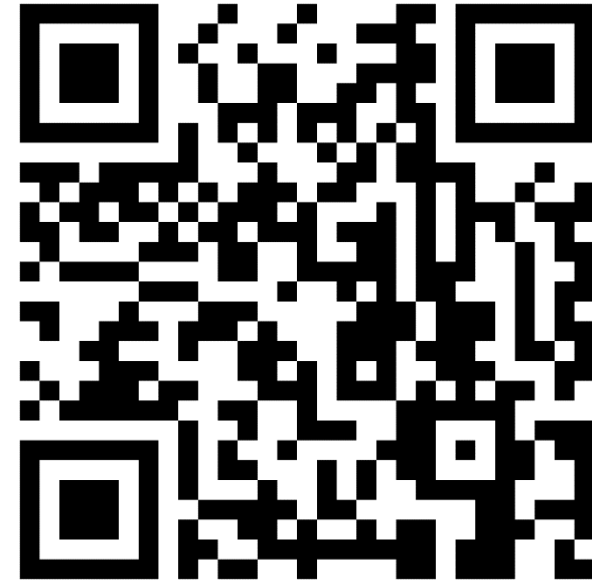
Magnetometers

- Determines your location with respect to Magnetic North (or South!)



6.3 Questions?

Mobile devices are equipped with sensors and hardware components. Discuss how mobile app could utilized these sensors or hardware components to provide a unique experience to users. Use a suitable example to explain your answer. (5 marks)



Conclusion

Explain basic requirements of a camera app

Identify and explain other hardware sensors

Develop app that could capture image and video

Implement app that can play audio and video files